CHUNYU ZHAO

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Research interest: Remote sensing and computer vision; multimodal data fusion; spatiotemporal reconstruction of remote sensing time series.

RESEARCH APPOINTMENT

Peking University | College of Environmental Sciences and Engineering **07.2026–Present**

Research Assistant | Supervisor: Dr. Jing Cheng (chengj10@pku.edu.cn)

- Produce and validate a global 1 km crop product suite covering cropland fraction, water-management regime, cropping intensity, crop type, harvested area, yield, and production.
- Investigate spatiotemporal dynamics of cropland use and their responses to environmental change.

EDUCATION

Beijing Normal University | State Key Laboratory of Remote Sensing and Digital Earth **07.2023–06.2026**

Master's degree in Remote Sensing Science and Technology | Supervisor: Prof. Zhiqiang Xiao (zhqxiao@bnu.edu.cn)

- GPA: 3.6/4.0 | Merit-based direct admission.
- Member of the Multiscale Satellite Remote Sensing (MUSES) R&D team (https://muses.bnu.edu.cn/)

Courses: Optical remote sensing; microwave remote sensing; remote sensing inversion and data assimilation method; remote sensing image processing and practice; remote sensing applications; principles and methods of remote sensing; principles and methods of geographic information system;

China University of Geosciences (Beijing) | School of Earth Sciences and Resources **07.2019–06.2023**

Bachelor's degree in Resources Prospecting Engineering | Thesis advisor: Prof. Gongwen Wang (gwwang@cugb.edu.cn)

- WAM: 86.84/100 | GPA: 3.43/4.0 | top 10%.
- Outstanding graduate of CUGB

Courses: Advanced mathematics surveying; computer language programming (c++); college chemistry; college physics; geosciences; remote sensing geology;

PUBLICATIONS

Peer-reviewed journal articles

1. Wang, A., Xiao, Z.*, **Zhao, C.**, Li, J., Zhang, Y., Song, J., & Yang, H. (2026). An Enhanced CycleGAN to Derive Temporally Continuous NDVI from Sentinel-1 SAR Images. *Remote Sensing*, 18(1), 56. ***(JCR Q2, CAS B2, IF=4.3).***
2. **Zhao, C.**, Xiao, Z., Zhang, Y., Yuan, C., & Yang, J.* (2025). Alteration Mineral Information Extraction Based on Image Super-Resolution Technology. *International Journal of Applied Earth Observation and Geoinformation,* 144, 104872. ***(JCR Q1, CAS B1 Top, IF=8.2, first author).***
3. **Zhao, C.**, Liu, Y.*, & Zhang, X. (2023). Application of Analytic Hierarchy Process in Optimal Allocation of Colorado River Under Drought Background. *Shaanxi Water Resources*, 2023(3); 26-29. *(In Chinese, first author).*

Submitted and in-preparation manuscripts

1. **Zhao, C.**, Wang, A., Zhou, X., Xiao, Z., Yang, H., & Song, J. (2026). Mapping Annual Dynamics of Aquatic Macrophytes in Baiyangdian wetland (2000-2024) Using Landsat Time Series and Deep Learning. ***(Submitted, first author).***
2. **Zhao, C.**, Xiao, Z., Zhang, Y., Song, J., & Yang, H. Temporal Scale Extension of LAI Products Derived from Satellite Observations Using Deep Neural Networks. *(In preparation).*

Conference paper

1. **Zhao, C.**, Xiao, Z.*, Song, J., Yang, H., & Zhang, Y. (2025). Time-Scale Extension Algorithm of MUSES LAI Products Using Image Super-Resolution. *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Brisbane, Australia.

PRESENTATIONS

Time-Scale Extension Algorithm of MUSES LAI Products Using Image Super-Resolution		**Oral**
1. The 45th IEEE International Geoscience and Remote Sensing Symposium	Brisbane, Australia	08.2025
2. The 7th National Forum on Quantitative Remote Sensing	Changchun, China	07.2025

RESEARCH EXPERIENCES

1. NSFC project **Participant** **01.2025–06.2026**

Integrated Remote Sensing Inversion and Spatiotemporal Extension of Vegetation Feature Parameters

- Develop spatial downscaling algorithms using deep learning and image super-resolution to enhance the spatiotemporal consistency of global remote sensing parameter products.
- Design parameter inversion algorithms to generate global remote sensing parameter products.

2. Geological Environment Monitoring and Forecasting **Participant** **05.2024–12.2024**

Research on Remote Sensing Assessment Methods for Water Conservation Functions of Ecosystems and Analysis of Spatial and Temporal Distribution Changes

- Conducted ecological field surveys in Baiyangdian Wetland, recording over 200 GPS locations and collecting category information at more than 20 key sampling sites.
- Analyzed vegetation distribution and temporal trends over 25 years; co-authored the Ecological Survey Report on Baiyangdian wetland.

3. Undergraduate thesis, CUGB **Leader** **01.2023–07.2023**

Remote sensing alteration extraction and mineralization prediction in Zhenghe Xiangluping, Fujian

- Conducted portable X-ray diffraction analyses on nearly 200 ore samples to characterize elemental compositions and mineralogical phases.
- Delineated five major and three secondary mineralization targets in the Xiangluping mining area.

ACADEMIC COMPETITION

1. Student Innovation and Entrepreneurship Training Program **S202211415201** **Leader** **01.2022–12.2022**

Remote Sensing of Porphyry Copper Alteration using ASTER Data **Municipal Level**

- Led more than 10 coordination meetings. The municipal-level project, one of four awarded at CUGB, was successfully completed.
- Processed ASTER data in ENVI 5.6 to extract hydrothermal alteration minerals in Duolong, Tibet; performed mapping and spatial analysis in ArcMap 10.8.

Supported by the National Key R&D Program: Global Paleogeographic Reconstruction and Deep-Time Big Data Key Technologies (2018YFE0204204).

2. Student Innovation and Entrepreneurship Training Program **X202111415269** **Leader** **01.2021–12.2021**

Microwave Synthesis of Perovskite Quantum Dots and Morphology Control Strategies **University Level**

- Optimized a single-parameter experiment; identified a 10% perovskite quantum dot concentration in the antisolvent as optimal for photoelectric performance.

HONORS and AWARDS

First-class Academic Scholarship (2023) BNU

Second-class Academic Scholarship (2019–2022, 2025) BNU, CUGB

Third Prize 17th Tsinghua National Environmental Technology Competition (North China Division) CUGB

4th Energy Conservation and Emission Reduction Competition;

5th “Green+” Innovation Competition.

LANGUAGE and SKILLS

English: **IELTS 6.5 overall** (Listening 6.0, Reading 7.0, Writing 6.0, Speaking 6.0);

Tools: Python, MATLAB, ArcGIS, ENVI, Adobe Illustrator.

RESEARCH INTERESTS

My research lies at the intersection of remote sensing and computer vision, with a focus on image restoration, multimodal data fusion, and deep learning for the spatiotemporal reconstruction of time series. My current work focuses on integrating multiple data to develop a suite of fine-resolution global agricultural data products spanning nearly four decades. These products aim to characterize changes in cropland extent and use, irrigation and water-management practices, cropping intensity, crop types and their spatial distribution, sown and harvested areas, crop yields, and total production. By capturing the long-term dynamics of these interconnected aspects of agricultural systems, I seek to understand how cropland use and agricultural management have evolved over time and how these changes relate to agricultural productivity and environmental change.